Generalizations of the Kelly Criterion

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1 Three Outcome Kelly Criterion

1.1 Introduction

The classical Kelly criterion maximizes the expected logarithmic growth of wealth for a binary bet. Here we consider a single bet of size f on an event with three mutually exclusive outcomes:

- **Outcome 1:** With probability p_1, the bet yields a profit of $b_1 f$ (i.e., wealth becomes $1+b_1 f$).
- **Outcome 2:** With probability p_2, the bet yields a profit of $b_2 f$ (i.e., wealth becomes $1+b_2 f$).
- **Outcome 0 (Loss):** With probability $1-p_1-p_2$, the bet loses the stake (i.e., wealth becomes $1-f$).

Our goal is to choose the optimal (normalized) fraction f^* of our bankroll that maximizes the expected long-term logarithm of wealth.

1.2 Expected Logarithmic Wealth

The expected logarithmic wealth $E(f)$ is given by

$$E(f) = p_1 \ln(1 + b_1 f) + p_2 \ln(1 + b_2 f) + (1 - p_1 - p_2) \ln(1 - f) \tag{1}$$

1.2.1 Domain

All logarithms must be defined, hence the admissible set for f is

$$\mathcal{D} = \Big\{ f : \; 1 - f > 0, \; 1 + b_1 f > 0, \; 1 + b_2 f > 0 \Big\} = \Big(\max\{-1/b_1, -1/b_2\}, \, 1 \Big)$$

1.2.2 Concavity

The second derivative

$$E''(f) = -\frac{p_1 b_1^2}{(1 + b_1 f)^2} - \frac{p_2 b_2^2}{(1 + b_2 f)^2} - \frac{1 - p_1 - p_2}{(1 - f)^2}$$

is strictly negative for all $f \in \mathcal{D}$, so E is strictly concave on $\mathcal{D}$.

1.3 Optimization

To maximize $E(f)$, we differentiate with respect to f and set the derivative equal to zero:

$$\frac{dE}{df} = \frac{p_1 b_1}{1 + b_1 f} + \frac{p_2 b_2}{1 + b_2 f} - \frac{1 - p_1 - p_2}{1 - f} = 0 \tag{2}$$

This first-order condition can be rewritten as:

$$\frac{p_1 b_1}{1 + b_1 f^*} + \frac{p_2 b_2}{1 + b_2 f^*} = \frac{1 - p_1 - p_2}{1 - f^*} \tag{3}$$

Note that we can generalize further here, obtaining the optimality condition for N outcomes:

$$\sum_{i=1}^{N} \frac{p_i b_i}{1 + b_i f^*} = \frac{1 - \sum_{i=1}^{N} p_i}{1 - f^*} \tag{4}$$

1.4 Derivation of the Quadratic Equation

A standard technique to solve (3) is to clear denominators. Multiplying both sides by the product $(1 + b_1 f)(1 + b_2 f)(1 - f)$ leads (after some algebra) to a quadratic equation in f of the form:

$$b_1 b_2 \, f^2 - [p_1 b_2 (b_1 + 1) + p_2 b_1 (b_2 + 1) - (b_1 + b_2)] \, f - [p_1 (b_1 + 1) + p_2 (b_2 + 1) - 1] = 0. \tag{5}$$

1.5 Final Result

The quadratic in Equation (5) can be solved using the quadratic formula. If we define:

$$B = p_1 b_2 (b_1 + 1) + p_2 b_1 (b_2 + 1) - (b_1 + b_2) \tag{6}$$

$$C = p_1 (b_1 + 1) + p_2 (b_2 + 1) - 1, \tag{7}$$

We can denote the root of (5) as

$$f_+ \;=\; \frac{B + \sqrt{B^2 + 4 b_1 b_2 C}}{2 b_1 b_2}$$

And because E is strictly concave on $\mathcal{D}$, the optimizer is

$$f^\star \;=\; \max\{0, \min\{\, 1 - \varepsilon \,, f_+\}\}$$

provided $f_+ \in \mathcal{D}$; otherwise select the feasible root in $\mathcal{D}$, or $f^\star = 0$ (no bet) if neither root is feasible.

1.6 Remarks

- **Consistency:** When $p_2 = 0$, Equation (**??**) reduces to the classical Kelly formula for a binary bet:
$$f^* = \frac{p_1 (b_1 + 1) - 1}{b_1}$$
- **Interpretation:** The first-order condition in (3) equates the marginal benefit (weighted by the probabilities and payoffs of winning outcomes) to the marginal cost (the risk of losing the wager). The positive solution of the quadratic provides the unique fraction f^* that maximizes the expected logarithmic growth.

2 Kelly Criterion with Ties

2.1 Introduction

When a bet may end in a tie, the outcome space consists of three mutually exclusive events:

- **Win:** With probability p_1, the bet wins and yields a profit of $b_1 f$ (wealth becomes $1 + b_1 f$).
- **Tie:** With probability p_2, the bet ends in a tie, and the wager is returned (wealth remains 1).
- **Loss:** With probability $1 - p_1 - p_2$, the bet loses the stake (wealth becomes $1 - f$).

Because the tie outcome yields zero profit ($\ln(1) = 0$), its only effect is to reduce the overall probability of a win. In what follows we derive the optimal betting fraction f^* by maximizing the expected logarithmic wealth.

2.2 Expected Logarithmic Wealth

Since a tie leaves wealth unchanged, the expected logarithmic wealth is given by

$$E(f) = p_1 \ln\Big(1 + b_1 f\Big) + p_2 \ln(1) + (1 - p_1 - p_2) \ln\Big(1 - f\Big)$$

Because $\ln(1) = 0$, this simplifies to

$$E(f) = p_1 \ln\Big(1 + b_1 f\Big) + (1 - p_1 - p_2) \ln\Big(1 - f\Big) \tag{8}$$

2.3 Optimization

To find the optimal fraction f^*, we differentiate $E(f)$ with respect to f:

$$\frac{dE}{df} = \frac{p_1 b_1}{1 + b_1 f} - \frac{1 - p_1 - p_2}{1 - f}$$

Setting the derivative equal to zero gives the first–order condition:

$$\frac{p_1 b_1}{1 + b_1 f^*} = \frac{1 - p_1 - p_2}{1 - f^*} \tag{9}$$

2.4 Derivation of the Optimal Bet Size

We now solve (9) for f^* Cross–multiplying, we obtain:

$$p_1 b_1 (1 - f^*) = (1 - p_1 - p_2)(1 + b_1 f^*)$$

Expanding both sides:

$$p_1 b_1 - p_1 b_1 f^* = (1 - p_1 - p_2) + (1 - p_1 - p_2) b_1 f^*$$

Collecting the terms involving f^* on one side:

$$p_1 b_1 - (1 - p_1 - p_2) = p_1 b_1 f^* + (1 - p_1 - p_2) b_1 f^*$$

Factor $b_1 f^*$ on the right:

$$p_1 b_1 - (1 - p_1 - p_2) = b_1 f^* \Big[p_1 + (1 - p_1 - p_2) \Big]$$

Notice that

$$p_1 + (1 - p_1 - p_2) = 1 - p_2$$

Thus,

$$p_1 b_1 - (1 - p_1 - p_2) = b_1 f^* (1 - p_2)$$

Solving for f^* yields:

$$f^* = \frac{p_1 b_1 - (1 - p_1 - p_2)}{b_1 (1 - p_2)} \tag{10}$$

2.5 Feasible Domain and Boundary Cases

The admissible set is $\mathcal{D} = \{f : 1 - f > 0,\ 1 + b_1 f > 0\} = (-1/b_1,\, 1)$. Since

$$E''(f) = -\frac{p_1 b_1^2}{(1 + b_1 f)^2} - \frac{1 - p_1 - p_2}{(1 - f)^2} < 0 \quad \text{on } \mathcal{D},$$

E is strictly concave and the solution in (10) is the unique interior maximizer when it lies in $\mathcal{D}$. Otherwise, the maximizer is attained at the boundary, practically implemented as

$$f^\star \leftarrow \max\{0, \min\{\, 1 - \varepsilon\,,\ \frac{p_1 b_1 - (1 - p_1 - p_2)}{b_1 (1 - p_2)}\ \}\}$$

If the numerator $p_1 b_1 - (1 - p_1 - p_2) \leq 0$, the optimal decision is $f^\star = 0$ (no bet)

2.6 Remarks

- **Consistency:** When the probability of a tie is zero (i.e. $p_2 = 0$), (10) reduces to the classical binary Kelly formula:

$$f^* = \frac{p_1 b_1 - (1 - p_1)}{b_1}$$

3 Kelly for Multiple Mutually Exclusive Bets

3.1 Introduction

In many scenarios (e.g. horse racing) there are several mutually exclusive outcomes. Suppose for the k-th outcome:

- The probability of winning is p_k
- The total amount wagered on outcome k is B_k (with $\sum_i B_i$ being the total bets on the event)
- The payoff odds are Q_k

Define

$$\beta_k = \frac{B_k}{\sum_i B_i} = \frac{D}{1+Q_k},$$

where $D = 1 - t$ is the dividend rate after the track take (or tax) t. Note that the quantity

$$\frac{D}{\beta_k} = 1 + Q_k$$

represents the revenue rate when outcome k wins. Let f_k denote the fraction of the bettor's funds wagered on outcome k. We want to select an optimal subset of outcomes S^o on which betting is favorable, as well as the optimal bet fractions f_k^o.

3.2 Assumptions and Notation

Two equivalent parameterizations are used:

- **Pari–mutuel (pool) model.** Let B_k be the total pool staked on outcome k, $B = \sum_i B_i$ the event pool, and $t \in [0,1)$ the track take. The dividend rate is $D = 1 - t$. The per-unit payout when k wins is $D\,B/B_k$. Defining

  $$\beta_k := \frac{B_k}{B}, \qquad Q_k := \frac{D}{\beta_k} - 1,$$

  yields $1 + Q_k = D/\beta_k$.

- **Fixed–odds model.** If posted odds Q_k are primitive (not generated by a pool), set $D = 1$ and $\beta_k := 1/(1+Q_k)$ so that the algebra below still applies. In this case β_k is a risk-scaling coefficient, not a literal pool share.

All results in this section hold under either parameterization as long as $1+Q_k > 0$ (i.e., $\beta_k \in (0,1]$).

3.3 Optimal Outcome Selection Algorithm

The algorithm to determine the optimal set S^o is as follows:

1. **Calculate Expected Revenue Rates:** For each outcome i, define its expected revenue rate by

   $$er_i = \frac{D\,p_i}{\beta_i} = p_i\,(Q_i + 1)$$

2. **Order Outcomes:** Rearrange the outcomes so that

$$er_1 \geq er_2 \geq \cdots \geq er_N$$

3. **Initialization:** Set the candidate optimal set $S = \varnothing$ (the empty set), let $k = 1$, and define

$$R(S) = 1$$

4. **Iterative Inclusion:** For $k = 1, 2, \ldots, N$:

   - If
   $$er_k = \frac{D\, p_k}{\beta_k} > R(S)\,,$$
   then include the k-th outcome into the set:
   $$S \leftarrow S \cup \{k\}\,,$$
   and update the reserve rate by
   $$R(S) = \frac{D \sum_{j \notin S} p_j}{D - \sum_{j \in S} \beta_j}$$
   Increment k and repeat.
   - Otherwise, terminate the procedure and set
   $$S^o = S$$

3.4 Calculation of Optimal Betting Fractions

For outcomes included in the optimal set S^o, the optimal fraction f_i to wager on outcome i is given by

$$f_i = p_i - \beta_i \frac{\sum_{j \notin S^o} p_j}{D - \sum_{j \in S^o} \beta_j}$$

It can be shown that the reserve rate corresponding to the optimal set satisfies

$$R(S^o) = 1 - \sum_{i \in S^o} f_i$$

3.5 Introducing Ties

Consider the modification that each bet may result in one of three outcomes:

- **Win:** With probability $p_{1,i}$, outcome i wins and yields a net profit of Q_i (i.e. total return $1 + Q_i$).
- **Tie:** With probability $p_{2,i}$, outcome i ends in a tie so that the wager is returned (wealth remains unchanged).
- **Loss:** With probability $1 - p_{1,i} - p_{2,i}$, outcome i loses the wager.

For a single bet in isolation, the binary Kelly fraction (adjusted for ties) is given by:

$$f_i^* = \frac{p_{1,i}Q_i - \big(1 - p_{1,i} - p_{2,i}\big)}{Q_i\big(1 - p_{2,i}\big)}$$

To extend this to n mutually exclusive bets, define:

$$\hat{p}_i = \frac{p_{1,i}Q_i - \big(1 - p_{1,i} - p_{2,i}\big)}{Q_i\big(1 - p_{2,i}\big)},$$

which represents the effective edge for outcome i (its isolated optimal fraction). We also define the risk–scaling factor:

$$\beta_i = \frac{1}{1 + Q_i}$$

Then the effective revenue rate is

$$er_i = \frac{\hat{p}_i}{\beta_i} = \hat{p}_i\,(1 + Q_i)$$

3.6 Algorithm

The procedure to determine the optimal subset S^o of outcomes to bet on and their corresponding fractions is as follows:

1. **Order Outcomes:** Sort outcomes in descending order of er_i.

2. **Initialization:** Set the candidate set $S = \varnothing$ and the reserve rate $R(S) = 1$.

3. **Iterative Inclusion:** For each outcome k in the sorted order, compute

$$R(S) = \frac{\sum_{j \notin S} \hat{p}_j}{1 - \sum_{j \in S} \beta_j}$$

   If

$$er_k = \hat{p}_k(1 + Q_k) > R(S),$$

   include outcome k in S. Otherwise, stop the inclusion process, setting $S^o = S$.

4. **Optimal Fractions:** For outcomes $i \in S^o$, assign

$$f_i = \hat{p}_i - \beta_i \frac{\sum_{j \notin S^o} \hat{p}_j}{1 - \sum_{j \in S^o} \beta_j}\ ,$$

   and set $f_i = 0$ for outcomes not in S^o.

When $n = 1$, this procedure recovers the binary–tie Kelly fraction:

$$f_1^* = \frac{p_{1,1}Q_1 - \big(1 - p_{1,1} - p_{2,1}\big)}{Q_1\big(1 - p_{2,1}\big)}$$

3.7 Proof of Optimality via Karush–Kuhn–Tucker

Let $f_k \geq 0$ be the fraction staked on outcome k and $R := 1 - \sum_i f_i$ the reserve. If outcome k occurs, terminal wealth is

$$W_k = R + f_k(1 + Q_k) = R + \frac{f_k}{\beta_k} D$$

We maximize

$$\max_{f \geq 0,\ \sum_i f_i \leq 1} \ \Phi(f) := \sum_{k=1}^{N} p_k \log W_k$$

The Lagrangian with multiplier $\lambda \geq 0$ for $\sum_i f_i \leq 1$ and $\mu_k \geq 0$ for $f_k \geq 0$ is

$$\mathcal{L}(f, \lambda, \mu) = \sum_k p_k \log W_k + \lambda \left(1 - \sum_i f_i \right) + \sum_k \mu_k f_k.$$

For any i with $f_i > 0$ (hence $\mu_i = 0$), the KKT stationarity condition gives

$$\frac{\partial \mathcal{L}}{\partial f_i} = \frac{p_i}{W_i} \Big((1 + Q_i) - 1 \Big) - \lambda = \frac{p_i}{W_i} \Big(\frac{D}{\beta_i} - 1 \Big) - \lambda = 0$$

Using $W_i = R + D f_i / \beta_i$ and solving for f_i yields

$$f_i \ = \ p_i - \beta_i R, \qquad \text{for all } i \text{ with } f_i > 0$$

Let $S := \{ i : \ f_i > 0 \}$ be the active set. Summing f_i over S gives

$$\sum_{i \in S} f_i = \sum_{i \in S} p_i - R \sum_{i \in S} \beta_i.$$

Since $R = 1 - \sum_i f_i$, rearranging gives the reserve identity

$$R \ = \ \frac{\sum_{j \notin S} p_j}{1 - \sum_{i \in S} \beta_i}$$

Remark. If $1 - \sum_{i \in S} \beta_i \leq 0$ for a candidate S, the problem becomes infeasible for that S (no reserve). The greedy inclusion rule prevents this. If no index satisfies $er_i > R$ at the start, then $S^o = \varnothing$ and $f^* \equiv 0$ (do not bet).

4 Kelly for Game Trees with Fixed Bet

In many practical scenarios, the outcome of a bet may depend on a sequence of events or decisions. Such situations can be modeled by a decision tree whose leaf nodes represent final outcomes, each with an associated probability and payout (or loss).

4.1 Decision Tree Generalization

Suppose the decision tree has L leaf nodes. For each leaf node i:

- p_i is the probability of reaching leaf i (determined by the probabilities along the branch)
- n_i is the number of events along the path to leaf i
- At each event j along the branch to leaf i, the wealth is updated by a factor

  $$1 + f\,\gamma_{ij}\,,$$

  where γ_{ij} is an effective coefficient that encapsulates the payoff (or loss) and any wager-scaling at that event

Thus, the overall wealth multiplier for leaf i is:

$$R_i(f) = \prod_{j=1}^{n_i} \Big(1 + f\,\gamma_{ij}\Big) \tag{11}$$

The expected logarithmic wealth is then:

$$E(f) = \sum_{i=1}^{L} p_i\, \ln\big(R_i(f)\big) = \sum_{i=1}^{L} p_i \sum_{j=1}^{n_i} \ln\Big(1 + f\,\gamma_{ij}\Big) \tag{12}$$

Differentiating with respect to f, the first-order condition becomes:

$$\frac{dE}{df} = \sum_{i=1}^{L} p_i \sum_{j=1}^{n_i} \frac{\gamma_{ij}}{1 + f\,\gamma_{ij}} = 0 \tag{13}$$

4.2 Domain and Assumptions

The fixed–f model here assumes the same fraction f multiplies each per-event coefficient γ_{ij} along any realized path. Feasibility requires

$$1 + f\,\gamma_{ij} > 0 \quad \text{for all factors that can occur with nonzero probability,}$$

so the admissible set is $\mathcal{D} = \{f :\ \min_{i,j}(1 + f\gamma_{ij}) > 0\}$. On $\mathcal{D}$, the objective is strictly concave in f whenever some $\gamma_{ij} \neq 0$, implying a unique optimizer. If actions influence which γ_{ij} can occur, the full problem is a control problem over both actions and f; the dynamic-programming outline selects actions to maximize expected log growth and then solves (13) on the pruned tree.

4.3 Pruning Suboptimal Branches

In many decision trees, not all branches correspond to optimal decisions. Suppose that at some decision nodes multiple actions are available, but only one (or a subset) maximizes the expected logarithmic growth of wealth. In such cases, the suboptimal branches can be pruned, and the probabilities adjusted accordingly.

Let $\mathcal{L}^*$ be the set of leaf nodes corresponding to the optimal decisions. The adjusted (or conditional) probability for each remaining leaf $i \in \mathcal{L}^*$ is:

$$\tilde{p}_i = \frac{p_i}{\sum_{k \in \mathcal{L}^*} p_k} \tag{14}$$

The expected logarithmic wealth for the pruned tree is:

$$E(f) = \sum_{i \in \mathcal{L}^*} \tilde{p}_i \sum_{j=1}^{n_i} \ln\Big(1 + f\,\gamma_{ij}\Big) \tag{15}$$

The first-order condition for the pruned tree becomes:

$$\frac{dE}{df} = \sum_{i \in \mathcal{L}^*} \tilde{p}_i \sum_{j=1}^{n_i} \frac{\gamma_{ij}}{1 + f\,\gamma_{ij}} = 0 \tag{16}$$

Solving (16) yields the optimal betting fraction f^* for the strategy that follows the optimal decisions at each branch.

4.4 Optimal Strategy via Dynamic Programming

To derive an optimal strategy in the context of a decision tree, one must determine the best action at each decision node in order to maximize long-term expected logarithmic wealth. This can be accomplished using dynamic programming:

- **State and Decision:** At any decision node, denote the state by x. Suppose there are several actions $a \in \mathcal{A}(x)$ available at state x.
- **Downstream Value:** Let $V(x, a)$ denote the expected logarithmic wealth, or value, of taking action a at state x and then following the optimal strategy thereafter. If action a leads to a set of downstream leaf nodes $\mathcal{L}_a$ with probabilities $p_i^{(a)}$ and wealth multipliers $R_i(f)$, then

$$V(x, a) = \sum_{i \in \mathcal{L}_a} p_i^{(a)} \ln\big(R_i(f)\big)$$

- **Bellman Optimality:** The optimal value at state x is then given by:

$$V^*(x) = \max_{a \in \mathcal{A}(x)} V(x, a)$$

  The action achieving this maximum, say $a^*(x)$, is the optimal action at that node.